

Felipe Khoury Kalil

✉ kalilfelipe30@gmail.com | 📍 São Paulo, Brazil
🌐 LinkedIn | 🐙 GitHub portfolio

EDUCATION

Universidade de São Paulo (USP) **2026 – Dec 2027 (Expected)**
Master of Science (M.Sc.) in Physics São Paulo, SP, Brazil
Focus: Plasma Physics | Advisor: Prof. José Helder Facundo Severo

Universidade Federal de Minas Gerais (UFMG) **2022 – 2025**
Bachelor of Science (B.Sc.) in Physics Belo Horizonte, MG, Brazil
Weighted Average: 94.7/100

RESEARCH EXPERIENCE

Master's Researcher **Mar 2026 – Present**
Universidade de São Paulo (USP) São Paulo, SP, Brazil

- **Project:** “Development of TCABR’s new magnetic diagnostics system”
- Studying ideal MHD and M3D-C¹ to model tokamak plasma equilibria of the TCABR upgrade and define operational requirements for the new sensor array.

Undergraduate Student Researcher **May 2023 – Feb 2025**
Universidade Federal de Minas Gerais (UFMG) Belo Horizonte, MG, Brazil

- Fabricated carbon-based nanosensors for the detection of liquid contaminants in fuel.
- Developed a cheap electronic tongue system using the fabricated sensors and standard data analysis techniques.
- Achieved very cost-efficient sensitivity with the system, detecting low contaminant concentrations while still being very economical to produce.

PUBLICATIONS

- Sampaio, M. P.; Alvim, R. G.; **Kalil, F. K.**; Aguiar, M. C. O. & Agero, U. (2025). “Analysis of optical pattern formation on glass: exploring light phenomena in the International Physicists’ Tournament.” **European Journal of Physics**. DOI: 10.1088/1361-6404/ae155b

PRESENTATIONS & CONFERENCES

- **Poster Presentations:** “Electronic tongue: development and applications of a liquid nanosensor based on carbon nanomaterials” [in Portuguese]
Semana do Conhecimento, Universidade Federal de Minas Gerais, 2023, 2024, 2025.
* **Winner of the Academic Relevance Award across all three years.**
- **Oral Presentation (IPT):** Defended complex physics solutions to international panels during the 2024, 2025, and 2026 International Physicists’ Tournaments.

HONORS & AWARDS

International Physicists’ Tournament (IPT)

- **9th Place** (2026) | **10th Place** (2025) | **7th Place** (2024)

Brazilian Physicists’ Tournament (BPT)

- **1st Place** (2023, 2024, 2025): Team Captain (2025); Recognized as Best Moderator (2024, 2025); 2nd Best Opponent (2024); and Team MVP (2024).

Semana do Conhecimento UFMG

- **Academic Relevance Award** (2023, 2024, 2025): Awarded three consecutive years for excellence in undergraduate poster presentations on the electronic tongue system.

WORKSHOPS & ACADEMIC TRAINING

- **XXIII Jorge André Swieca Summer School on Particles and Fields** — Sociedade Brasileira de Física (Aug 2025)

STANDARDIZED TESTS

- **Brazilian Unified Physics Exam (EUF):** 98th Percentile **Oct 2025**
Note: This is the standardized exam used for both Master's and Ph.D. admissions across Brazilian institutions.

SKILLS

- **Analytical & Computational Modeling:** Mathematical Modeling, Multiphysics Simulation (COMSOL Multiphysics[®]), Numerical Methods.
- **Experimental & Data Analysis:** Instrumentation Prototyping, Signal Processing, Noisy Data Treatment, Data Visualization.
- **Programming & Software:** Python (NumPy, SciPy, Matplotlib, pandas, Keras), Machine Learning, L^AT_EX.
- **Languages:** Portuguese (Native), English (Fluent), Spanish (Intermediate), Italian (Basic).